

# A Bayesian Regression Study of Home Prices in Seattle, Washington

## 1. Introduction

Understanding the drivers of property prices is a central problem in real estate economics and in public policy. Home prices are influenced by structural characteristics, neighborhood quality, and locational attributes. Identifying the contributions of individual features, such as living area, construction quality, and waterfront access, is of importance to homeowners, and policymakers. This study applies Bayesian linear regression to the King County, Washington home sales dataset, which contains over 20,000 records of residential sales between May 2014 and May 2015.

Our primary inferential goal is to identify the largest and most significant predictors of home sale price, with uncertainty being quantified with posterior distributions. Rather than treating coefficients as point estimates, we will use their posterior distributions to assess each predictor's effect on log price. We will also examine the model's sensitivity to prior specification. Specifically, we would like to see whether a weakly informative prior or an informative prior centered on OLS estimates yield meaningfully different results. Finally, we will assess whether the inclusion of non-linear and interaction effects improves model performance on this dataset.

We begin our study with exploratory data analysis, including descriptive statistics, correlation structure, and a visualization of key predictors against the outcome. We then fit a baseline OLS regression to establish coefficient estimates and assess our model via residual diagnostics. The primary modeling stage fits two Bayesian multiple regression models with different priors using Stan. Finally, a BART model is fit as a nonparametric check to assess whether there is any non-linear structure to our data.

## 2. Data Collection and Description

The dataset contains home sale prices for King County, Washington (which includes Seattle), and includes homes sold between May 2014 and May 2015. The original source is King County's Public Property Records, and was compiled and documented by the University of Chicago's Center for Spatial Data Science (2020). The dataset contains 21,613 observations and 21 variables, which includes sale price (the outcome variable), structural characteristics (bedrooms, bathrooms, square footage), property attributes (waterfront, condition, grade), location (zip code, lat, long), and neighborhood metrics (average square footage of the 15 closest homes).

The 'id' and 'date' variables are excluded as they have no predictive value for price modeling. Similarly, 'lat' and 'long' are dropped as their relationship with price is spatial, making them unsuitable as linear predictors. The 'zipcode' variable is excluded due to its 70 unique values, which makes it a poor predictor without employing hierarchical modeling. Additionally, 'sqft\_above' and 'sqft\_basement' are

removed as they are collinear with `sqft_living` by construction. Finally, `yr_renovated` is excluded due to the extreme skew/large proportion of zeros.

Next, the `yr_built` variable is replaced by `age`, computed as  $2015 - \text{'yr\_built'}$ , where 2015 corresponds to the end of the data collection period. This transformation is preferred because a home's age has a more natural and interpretable relationship with price than its construction year. We are left with a total of 11 variables, containing a mix of continuous, discrete, ordinal, and binary predictors.

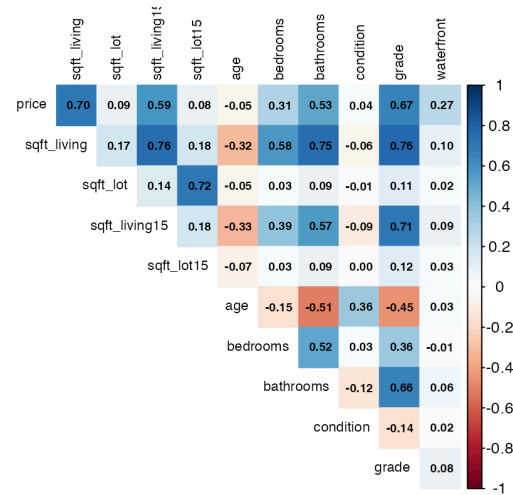
Turning to the summary statistics table (Figure 1), the median home in the dataset has 3 bedrooms, 2.25 bathrooms, and approximately 1,910 square feet of living space. The `sqft_lot` and `sqft_lot15` measures show extreme right skew, with their means roughly double their medians. This indicates that a small number of very large lots are pulling the distribution to the right. It is worth noting that `bedrooms` has a maximum of 33, which may be a data entry error.

The correlation matrix (Figure 2) highlights several relationships among the predictors. The most correlated pairs are `sqft_living` with `sqft_living15` (0.76) and `bathrooms`; (0.75), and `sqft_lot` with `sqft_lot15` (0.72). Conversely, `age` has a moderate negative correlation with `bathrooms` (-0.51) and `grade` (-0.45). While this level of multicollinearity would inflate standard errors and destabilize coefficient estimates under the frequentist approach, the Bayesian framework avoids this through prior regularization. The normal priors placed on each parameter act to stabilize the posterior distributions.

**Figure 1: Descriptive Statistics**

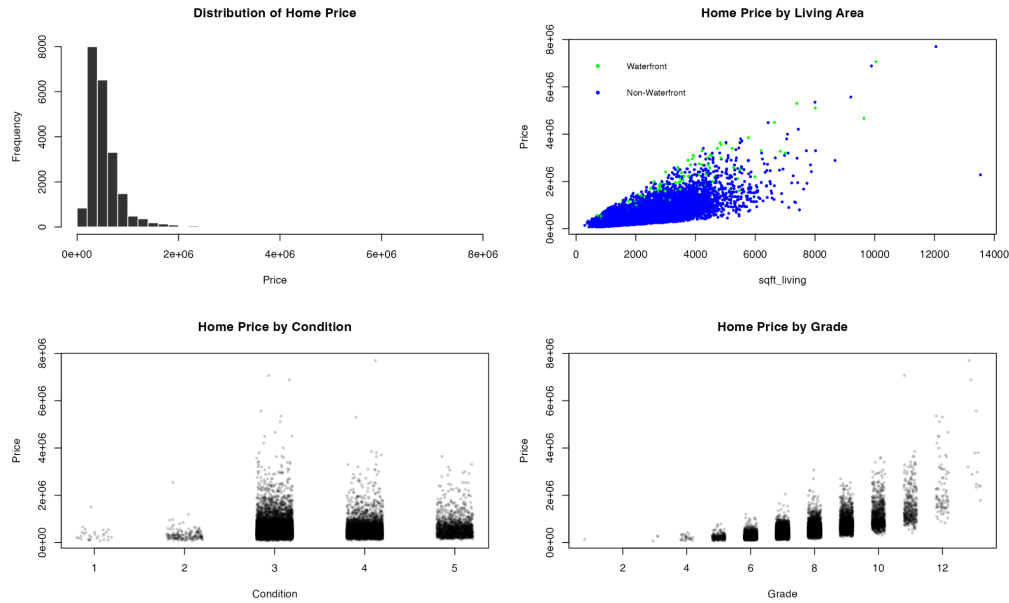
| Descriptive Statistics |          |           |           |        |              |
|------------------------|----------|-----------|-----------|--------|--------------|
| Variable               | Median   | Mean      | SD        | Min    | Max          |
| bedrooms               | 3.00     | 3.37      | 0.93      | 0.00   | 33.00        |
| bathrooms              | 2.25     | 2.11      | 0.77      | 0.00   | 8.00         |
| sqft_living            | 1,910.00 | 2,079.90  | 918.44    | 290.00 | 13,540.00    |
| sqft_lot               | 7,618.00 | 15,106.97 | 41,420.51 | 520.00 | 1,651,359.00 |
| age                    | 40.00    | 43.99     | 29.37     | 0.00   | 115.00       |
| sqft_living15          | 1,840.00 | 1,986.55  | 685.39    | 399.00 | 6,210.00     |
| sqft_lot15             | 7,620.00 | 12,768.46 | 27,304.18 | 651.00 | 871,200.00   |

**Figure 2: Correlation Matrix**



Looking at the exploratory plots (Figures 3-6), the price distribution shows a strong right skew with a long tail. In addition, the scatter plot exhibits heteroscedasticity: variance in price appears to be increasing with square footage. Both of these observations motivate a log transformation of the outcome variable. Grade shows a clear positive relationship with price, while condition shows a weaker relationship, which suggests grade is the more informative of the two ordinal predictors. There is a strong class imbalance for waterfront properties, but they appear to have higher prices and tend toward larger living areas.

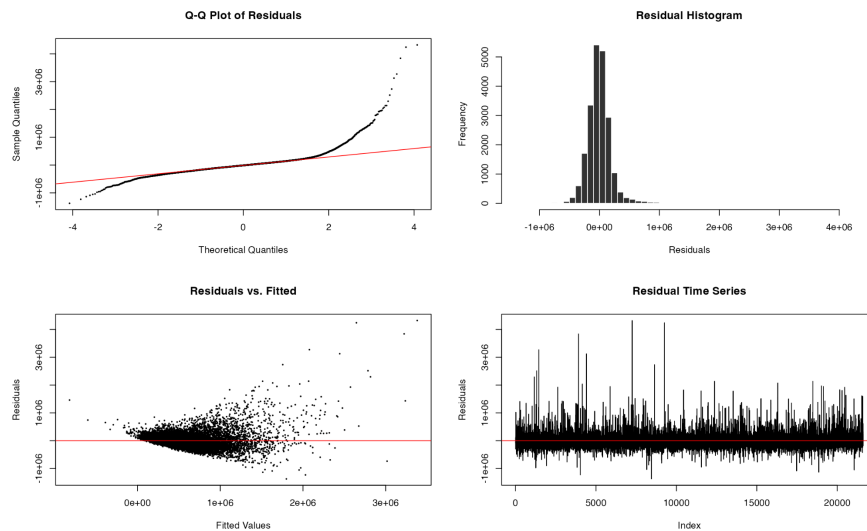
### Figures 3-6: Exploratory Plots



## 3. Statistical Methods

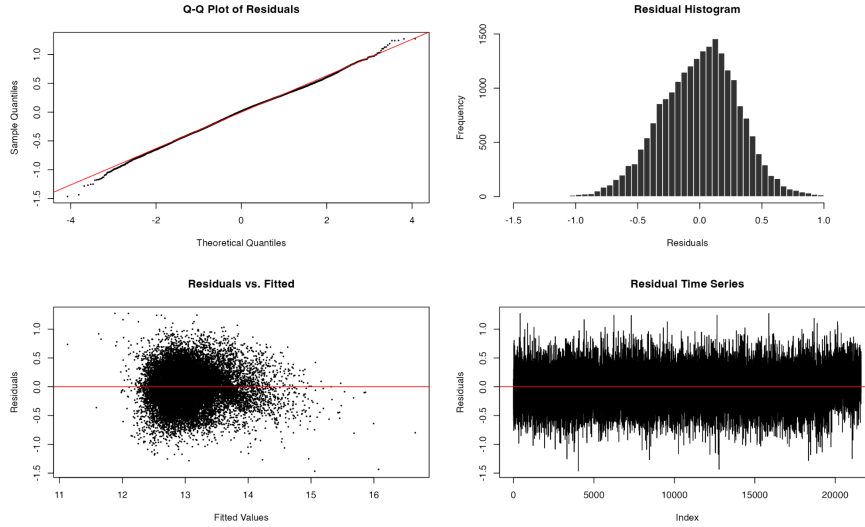
The analysis starts with an ordinary least squares (OLS) regression of raw home price on all predictors as a baseline. This will be useful later when constructing the Bayesian models. Diagnostic plots, including a histogram, QQ plot, residuals versus fitted values, and a residual time series, are then examined to assess normality and homoscedasticity. As seen in Figures 7-10, the residuals from the raw price model illustrate non-normality and heteroscedasticity. This motivates the use of a transformation on the outcome price.

### Figures 7-10: Regression Diagnostics (Raw Price)



The Box-Cox procedure is applied to identify the best power transformation, which estimates the parameter  $\lambda$ . The estimated  $\lambda \approx 0.10$  is close to zero, meaning a log transformation is appropriate. A second OLS regression is then fit with log price, and the same diagnostic plots are re-examined to confirm that the model assumptions are met. In Figures 11-14, we can see from the QQ plot and the histogram that residuals look normal, and the residuals versus fitted values no longer exhibit non-constant variance. Going forward, the Bayesian models will use log price as the outcome variable.

**Figure 11-14: Regression Diagnostics (Log Price)**



Two Bayesian multiple regression models are fit next. The two models differ only in their prior specifications. Both models take the form,

$$\log(\text{price}_i) = \alpha + x_i^T \beta + \epsilon_i$$

where all predictors are standardized to have a mean of zero and unit variance. The predictors are standardized because Stan's Hamiltonian Monte Carlo algorithm struggles to converge with the vastly different scales of the variables. As opposed to using the multivariate normal prior with covariance matrix  $\Sigma = \sigma^2(X^T X)^{-1}$ , independent normal priors are used for each  $\beta_i$ . Given the very large sample size, we suspect that the posterior distributions will be relatively invariant to prior specification. In addition, independent normal priors allow for greater computational efficiency.

The first model uses weakly informative priors. The alpha prior is a normal centered at zero with a standard deviation of 10, which is essentially flat over a large range of mean log price. Each beta prior is also distributed normally, and they are weakly informative on the standardized scale (standard deviation of 1). A coefficient larger than 1 or 2 in absolute value would imply a very large effect on log price, so this prior provides regularization toward zero but doesn't constrain the posterior. Sigma has a half-normal prior, which places most probability on small positive values while still remaining weakly informative. This follows the recommendation of Gelman (2006). The prior specification is below:

$$\begin{aligned}\alpha &\sim N(0, 10) \\ \beta_i &\sim N(0, 1) \\ \sigma &\sim HalfNormal(0, 1)\end{aligned}$$

The second model uses informative priors derived from the OLS estimates on the log price model. The motivation is that the OLS estimates provide a reliable starting point that can be used to center the priors. This could potentially improve sampling efficiency over the weakly informative model. The alpha prior uses the OLS intercept as its mean (adjusted for standardization), where the standard deviation is the OLS standard error of the intercept. Each beta coefficient has a prior centered on its OLS estimate (again, adjusted for standardization), where the standard deviation is the corresponding OLS standard error. Finally, the sigma prior uses the OLS residual standard error as the mean, and the scale parameter is set as the asymptotic standard error of the OLS model. The prior specification is below:

$$\begin{aligned}\alpha &\sim N(\hat{\alpha}, \hat{\sigma}_\alpha) \\ \beta_i &\sim N(\hat{\beta}_i, \hat{\sigma}_{\beta_i}) \\ \sigma &\sim HalfNormal(\hat{\sigma}, \hat{\sigma} / \sqrt{2(n - p - 1)})\end{aligned}$$

BART is fit as a nonparametric check for nonlinear and interaction effects that the linear specification may fail to capture. Unlike the Bayesian regression models, BART makes no assumptions about the functional form of the relationship between predictors and outcome. Rather than fitting a single global function, BART approximates  $f(x)$  as a sum of  $m$  regression trees, where each tree is fit to the residuals left by the others. This backfitting procedure allows the ensemble to capture nonlinear relationships and interaction effects automatically, without requiring one to specify them in advance.

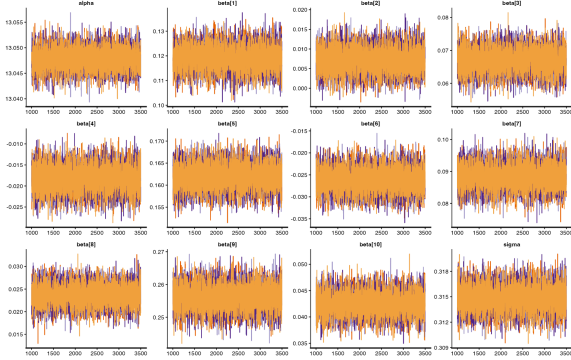
The prior on the tree structure is parameterized by base  $\alpha$  and power  $\beta = 2$ , which controls model complexity by making splits at deeper nodes increasingly unlikely, enforcing a preference for shallow, weak learners whose collective sum approximates the underlying function. The splitting variable and splitting rule priors are left at their noninformative defaults, placing uniform probability over all predictors and candidate split points. The base parameter was selected by evaluating a grid of values  $\alpha \in \{0.25, 0.50, 0.75, 0.95\}$  via 5-fold cross-validation, minimizing held-out RMSE, fully allowing the data to determine the level of tree complexity needed. The final model was refit on the full dataset using the selected base parameter. Comparing BART's fit to the Bayesian linear models provides a direct assessment of whether the linear assumption is adequate for this dataset.

## 4. Results

Both Bayesian multiple regression models were fit using Stan, which implements the Hamiltonian Monte Carlo algorithm to draw samples from the posterior distribution. Both models have excellent convergence and show strong sampling efficiency. The  $\hat{R}$  statistic is equal to 1 for all parameters across the two models, indicating that the four chains fully mixed and converged. The trace plots for all parameters confirm this visually, showing the "fuzzy caterpillar" pattern with no visible trends and consistency between chains.

The effective sample size (ESS) measures the number of independent draws from the posterior. Both models produce very high ESS values, from roughly 8,000 to 20,000. The weakly informative model achieves slightly higher ESS for most parameters, but both produce exceptional results and are sufficient for posterior inference. The  $\hat{R}$  statistics and ESS values confirm that both models have converged and that the posterior summaries are trustworthy.

**Fig. 15: Trace Plots (Weakly Informative)**



**Fig. 16: Trace Plots (Informative)**

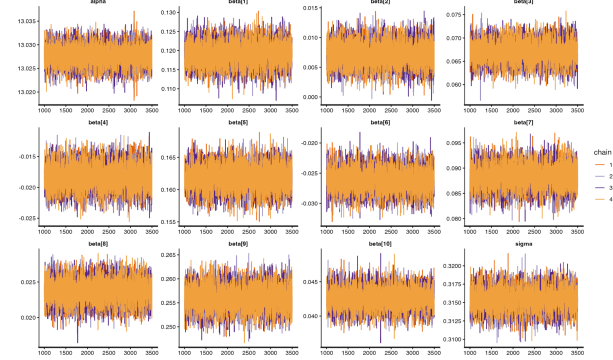


Figure 17 contains the prior and posterior densities for the parameters under the weakly informative prior model. For all parameters, the prior (red) appears nearly flat relative to the posterior (blue). This is because of scaling rather than the priors being truly uninformative. The  $N(0, 1)$  priors on the standardized betas and the  $N(0, 10)$  prior on alpha are plotted on the same y-axis scale as the very dense posteriors, making them appear diffuse by comparison. For alpha, the prior is quite wide, and thus the likelihood heavily influences the posterior. For the betas, the  $N(0, 1)$  prior does provide mild regularization by shrinking coefficients toward zero, but the very large sample size means the posteriors are driven almost entirely by the data. For sigma, the half-normal prior is also outweighed by the likelihood. Overall, these plots reinforce that with over 20,000 observations, the data dominates the posterior regardless of prior specification.

**Figure 17: Posteriors vs. Weakly Informative Priors**

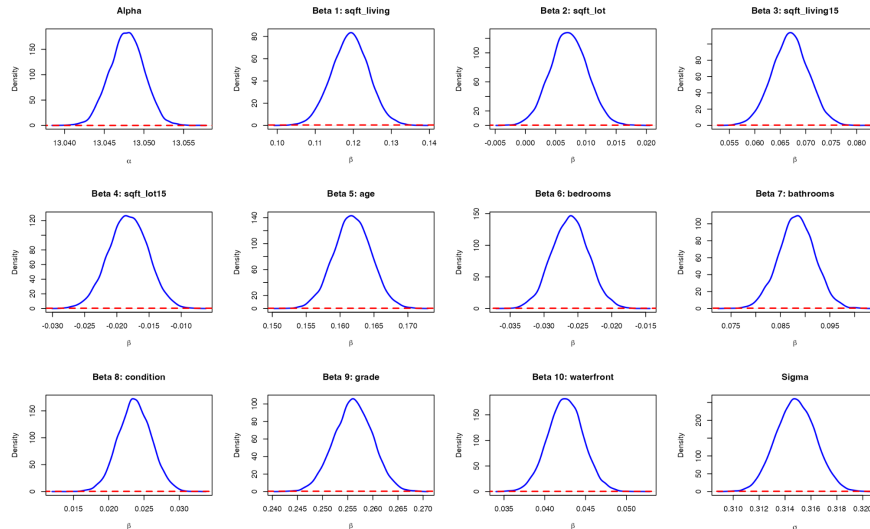
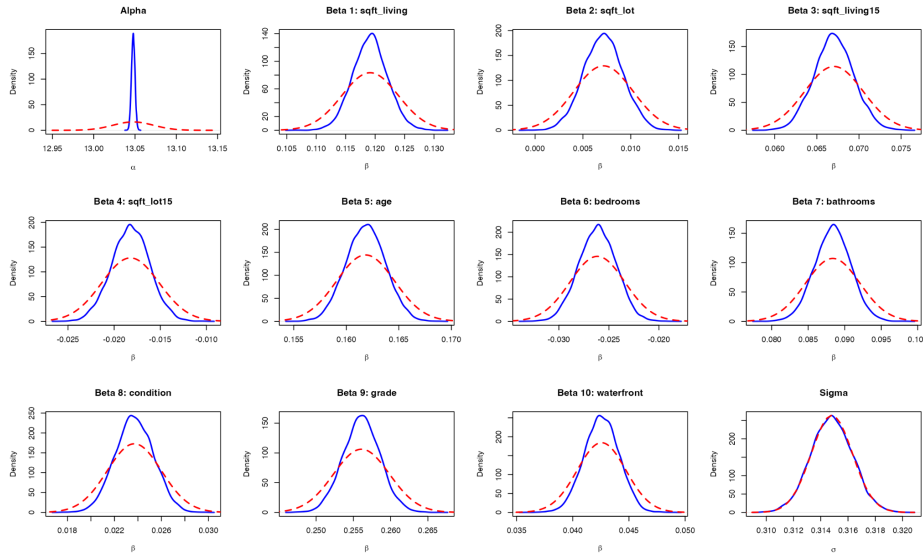


Figure 18 presents the prior and posterior densities for all parameters under the informative prior model. The posterior density (blue) of alpha is much tighter than the prior, with a sharp spike versus a wide curve. The likelihood completely dominates for the intercept. For our betas, the posteriors are consistently tighter than the prior (red) across all coefficients. However, the two densities are well-aligned in location. This confirms that the informative priors are good estimates and consistent with the data. For sigma, the prior and posterior are nearly identical, which reflects the fact that the standard error of sigma from the OLS regression is a very accurate approximation of the posterior uncertainty. The OLS asymptotic formula was used to calculate the standard error of sigma, and works well given the very large sample size. Overall, these plots illustrate that the likelihood dominates for all parameters, and the informative priors are consistent with the data.

**Figure 18: Posteriors vs. Informative Priors**



Figures 19 and 20 contain the posterior summary tables for the weakly informative and informative prior models, respectively. The posterior means are almost identical across both models, with differences no larger than 0.0001 for any coefficient. This confirms that the likelihood dominates the posterior and the choice of prior has little impact on the point estimates. However, the informative prior model does produce tighter 95% HPDIs across all coefficients, reflecting the additional information contributed by the OLS-derived priors. The most important predictors are grade ( $\beta = 0.256$ ), age ( $\beta = 0.162$ ), and sqft\_living ( $\beta = 0.119$ ), each with HPDIs that exclude zero by a wide margin. Notably, bedrooms ( $\beta = -0.026$ ) and sqft\_lot15 ( $\beta = -0.018$ ) are the only predictors with negative posterior means, with HPDIs entirely below zero. This indicates that these have negative effects conditional on the other predictors in the model. Every coefficient included in the models is statistically significant at the  $\alpha = 5\%$  significance level.

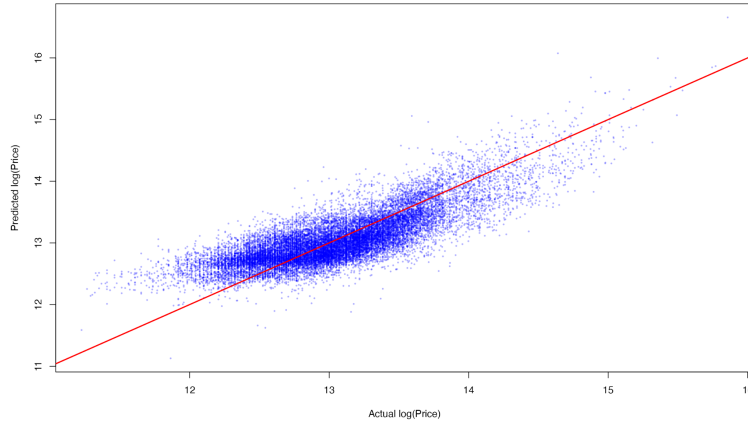
**Figure 19: Weakly Informative Priors**

| Parameter             | Mean    | 95% HPDI           |
|-----------------------|---------|--------------------|
| Beta 1: sqft_living   | 0.1192  | [0.1095, 0.1284]   |
| Beta 2: sqft_lot      | 0.0072  | [0.0011, 0.013]    |
| Beta 3: sqft_living15 | 0.067   | [0.0599, 0.0739]   |
| Beta 4: sqft_lot15    | -0.0183 | [-0.0241, -0.0122] |
| Beta 5: age           | 0.1618  | [0.1563, 0.1671]   |
| Beta 6: bedrooms      | -0.0261 | [-0.0315, -0.0208] |
| Beta 7: bathrooms     | 0.0883  | [0.081, 0.0953]    |
| Beta 8: condition     | 0.0237  | [0.0191, 0.0283]   |
| Beta 9: grade         | 0.2561  | [0.2486, 0.2634]   |
| Beta 10: waterfront   | 0.0426  | [0.0383, 0.0467]   |

**Figure 20: Informative Priors**

| Parameter             | Mean    | 95% HPDI           |
|-----------------------|---------|--------------------|
| Beta 1: sqft_living   | 0.1191  | [0.1136, 0.1252]   |
| Beta 2: sqft_lot      | 0.0071  | [0.0033, 0.0113]   |
| Beta 3: sqft_living15 | 0.067   | [0.0623, 0.0715]   |
| Beta 4: sqft_lot15    | -0.0182 | [-0.0224, -0.0142] |
| Beta 5: age           | 0.1618  | [0.1581, 0.1654]   |
| Beta 6: bedrooms      | -0.0261 | [-0.0298, -0.0226] |
| Beta 7: bathrooms     | 0.0884  | [0.0836, 0.0933]   |
| Beta 8: condition     | 0.0237  | [0.0206, 0.0269]   |
| Beta 9: grade         | 0.2561  | [0.251, 0.2608]    |
| Beta 10: waterfront   | 0.0426  | [0.0396, 0.0455]   |

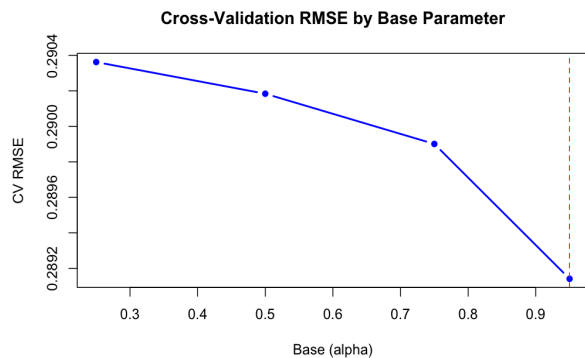
Figure 21 shows the posterior predictive log price against the actual log price for all 21,613 observations, derived from the informative prior model. The red line represents perfect prediction, with points lying above the line indicating overprediction and points below the line indicating underprediction. The clustering of points around the diagonal across the middle of the plot illustrates that the model fits well for average homes, but it is less accurate at the two extremes. It is worth noting that some curvature/nonlinearity exists. This is the kind of signal that a BART model is well-suited for. It may better fit the data in the tails, and if so, it would suggest nonlinearity or interaction effects are driving the remaining structure.

**Figure 21: Final Fitted Plot (Informative Prior)**

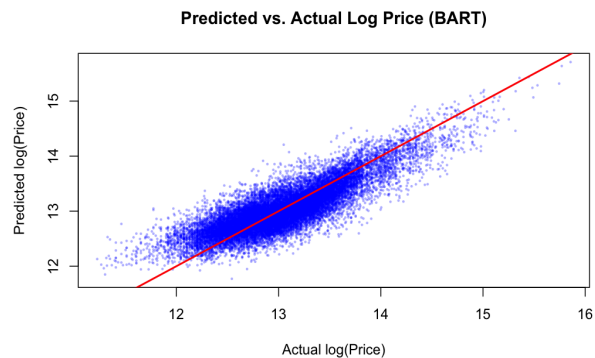
The BART model was fitted using 5-fold cross-validation over a grid of base values  $\alpha \in \{0.25, 0.50, 0.75, 0.95\}$ , with all other parameters held fixed at 'ntree' = 200. The CV RMSE decreased monotonically with  $\alpha$ , from 0.2904 at  $\alpha = 0.25$  to 0.2891 at  $\alpha = 0.95$ , and the optimal base was selected as 0.95 (Figure 22). The monotonic decrease across the full grid indicates that the data genuinely benefits from deeper, more complex trees, consistent with the presence of nonlinear structure. The final model was refitted on the full dataset using base = 0.95, achieving an in-sample RMSE of 0.2806 compared to 0.3147 for the OLS log price model, an improvement of approximately 11%.



**Figure 22: Cross-Validation RMSE by Base Parameter**

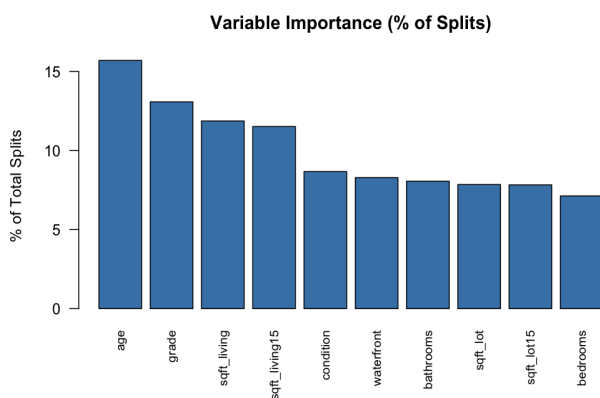


**Figure 23: Final Fitted Plot (BART)**

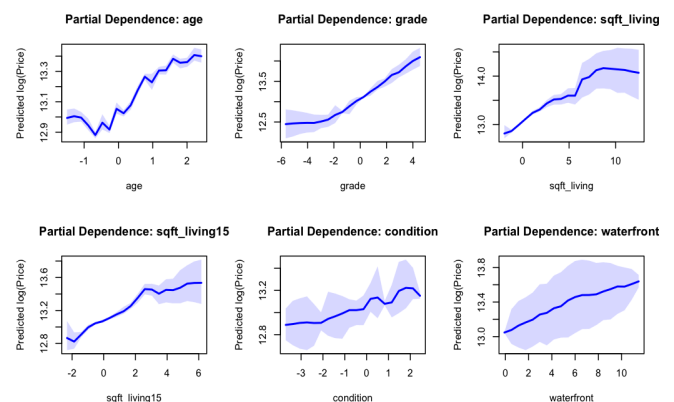


Comparing Figure 23 to Figure 21, the systematic bowing pattern in the Bayesian linear model's fitted plot is substantially reduced under BART, with predictions clustering more uniformly around the diagonal across the full price range. The variable importance plot (Figure 24) broadly agrees with the Bayesian regression findings: age, grade, and sqft\_living are the three most frequently split predictors, though BART ranks age above grade, suggesting age has a stronger nonlinear relationship with price than its linear coefficient alone captured. The partial dependence plots (Figure 25) confirm this: age exhibits a nonlinear relationship with price, dipping for recently built homes before increasing steadily for older homes, likely reflecting the premium placed on established neighborhoods. Grade and sqft\_living are approximately monotone but show diminishing returns at higher values. Condition remains flat and uncertain, consistent with its weak signal in the Bayesian regression. Taken together, these results confirm that while the linear model identified the correct predictors, it understated the complexity of their relationships with price.

**Figure 24: Variable Importance (% of Splits)**



**Figure 25: Partial Dependence Plots**



## 5. Conclusion

This project applied Bayesian multiple regression to the King County, Washington home sales dataset to investigate the primary drivers of home price. After exploratory data analysis, a baseline OLS regression was fit. Next, a Box-Cox procedure was run, which motivated a log transformation of the outcome variable. Two Bayesian multiple regression models were then fit with Stan, one with weakly informative priors and one with informative priors derived from the OLS estimates. All predictors were standardized prior to model fitting. Both models had excellent convergence, with  $\hat{R} = 1$  and high effective sample sizes for all parameters.

The results show that construction quality and home size are the primary drivers of home price. On the standardized log price scale, grade was the most influential predictor with  $\beta = 0.256$  and a 95% HPDI of  $[0.251, 0.261]$ . This means that a one standard deviation increase in construction grade is associated with a 29.2% increase in price ( $e^{0.256} - 1$ ). Age is the second strongest predictor ( $\beta = 0.162$ ), reflecting a positive price effect for older homes after controlling for other variables. Next, `sqft_living` ( $\beta = 0.119$ ) and `sqft_living15` ( $\beta = 0.067$ ) illustrate that both a home's size and the size of neighboring homes positively influence price. Interestingly, `bedrooms` ( $\beta = -0.026$ ) has a negative effect conditional on square footage and the other covariates. All predictors have 95% HPDIs that exclude zero, meaning that each is a statistically significant predictor of home price at the  $\alpha = 5\%$  significance level.

The BART model confirmed that the linear specification, while identifying the correct predictors, understated the complexity of their relationships with price. Cross-validation selected `base = 0.95`, the most complex configuration in the grid, and BART achieved an in-sample RMSE of 0.2806 compared to 0.3147 for OLS, an 11% improvement. The fitted plot showed substantially reduced systematic bias relative to the Bayesian linear model, particularly in the tails of the price distribution. Partial dependence plots revealed that age has a nonlinear relationship with price, with a U-shaped pattern across the range of home ages, and that grade and `sqft_living` exhibit diminishing returns at higher values. Variable importance was broadly consistent with the Bayesian multiple regression findings, with age, grade, and `sqft_living` ranking as the three most frequently split predictors. Together, these results suggest that a nonparametric model better captures the true structure of the data, and that the linear model's coefficients should be interpreted as approximations of relationships that are in reality more complex.

Future work could extend this analysis by incorporating a hierarchical model with zip code as a grouping variable, further exploring the nonlinear and interaction effects identified by the BART model, and applying the model to more recent home sales data to assess whether the predictors are consistent over time.

# Appendix

## I. Software Output

### OLS Regression (Raw Price)

```
Residuals:
    Min       1Q   Median       3Q      Max
-1374309 -112903  -10898    91100  4319676

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -1.018e+06  1.698e+04 -59.964 < 2e-16 ***
sqft_living  1.666e+02  3.607e+00  46.170 < 2e-16 ***
sqft_lot     8.813e-03  5.170e-02   0.170  0.865
sqft_living15 2.846e+01  3.531e+00  8.058 8.17e-16 ***
sqft_lot15   -5.809e-01  7.902e-02 -7.351 2.03e-13 ***
age          3.748e+03  6.526e+01  57.422 < 2e-16 ***
bedrooms     -4.317e+04  2.038e+03 -21.185 < 2e-16 ***
bathrooms    5.637e+04  3.352e+03  16.815 < 2e-16 ***
condition    1.716e+04  2.460e+03  6.976 3.12e-12 ***
grade        1.254e+05  2.214e+03  56.633 < 2e-16 ***
waterfront   7.213e+05  1.736e+04  41.554 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

### OLS Regression (Log Price)

```
Residuals:
    Min       1Q   Median       3Q      Max
-1.46620 -0.21276  0.01737  0.21374  1.27507

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.040e+01  2.450e-02  424.711 < 2e-16 ***
sqft_living  1.297e-04  5.203e-06  24.933 < 2e-16 ***
sqft_lot     1.726e-07  7.457e-08   2.314  0.0207 *
sqft_living15 9.779e-05  5.094e-06  19.200 < 2e-16 ***
sqft_lot15   -6.677e-07  1.140e-07 -5.859 4.72e-09 ***
age          5.510e-03  9.413e-05  58.530 < 2e-16 ***
bedrooms     -2.812e-02  2.939e-03 -9.566 < 2e-16 ***
bathrooms    1.147e-01  4.835e-03  23.729 < 2e-16 ***
condition    3.642e-02  3.549e-03  10.262 < 2e-16 ***
grade        2.179e-01  3.194e-03  68.209 < 2e-16 ***
waterfront   4.919e-01  2.504e-02  19.649 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

### Bayesian Regression (Weakly Informative)

4 chains, each with iter=3500; warmup=1000; thin=1;  
post-warmup draws per chain=2500, total post-warmup draws=10000.

|          | mean  | se_mean | sd | 2.5%  | 25%   | 50%   | 75%   | 97.5% | n_eff | Rhat |
|----------|-------|---------|----|-------|-------|-------|-------|-------|-------|------|
| alpha    | 13.05 | 0       | 0  | 13.04 | 13.05 | 13.05 | 13.05 | 13.05 | 16239 | 1    |
| beta[1]  | 0.12  | 0       | 0  | 0.11  | 0.12  | 0.12  | 0.12  | 0.13  | 7968  | 1    |
| beta[2]  | 0.01  | 0       | 0  | 0.00  | 0.01  | 0.01  | 0.01  | 0.01  | 10448 | 1    |
| beta[3]  | 0.07  | 0       | 0  | 0.06  | 0.06  | 0.07  | 0.07  | 0.07  | 10445 | 1    |
| beta[4]  | -0.02 | 0       | 0  | -0.02 | -0.02 | -0.02 | -0.02 | -0.01 | 10239 | 1    |
| beta[5]  | 0.16  | 0       | 0  | 0.16  | 0.16  | 0.16  | 0.16  | 0.17  | 10199 | 1    |
| beta[6]  | -0.03 | 0       | 0  | -0.03 | -0.03 | -0.03 | -0.02 | -0.02 | 12862 | 1    |
| beta[7]  | 0.09  | 0       | 0  | 0.08  | 0.09  | 0.09  | 0.09  | 0.10  | 9878  | 1    |
| beta[8]  | 0.02  | 0       | 0  | 0.02  | 0.02  | 0.02  | 0.03  | 0.03  | 15381 | 1    |
| beta[9]  | 0.26  | 0       | 0  | 0.25  | 0.25  | 0.26  | 0.26  | 0.26  | 10455 | 1    |
| beta[10] | 0.04  | 0       | 0  | 0.04  | 0.04  | 0.04  | 0.04  | 0.05  | 14735 | 1    |
| sigma    | 0.31  | 0       | 0  | 0.31  | 0.31  | 0.31  | 0.32  | 0.32  | 11206 | 1    |

### Bayesian Regression (Informative)

4 chains, each with iter=3500; warmup=1000; thin=1;  
post-warmup draws per chain=2500, total post-warmup draws=10000.

|          | mean  | se_mean | sd | 2.5%  | 25%   | 50%   | 75%   | 97.5% | n_eff | Rhat |
|----------|-------|---------|----|-------|-------|-------|-------|-------|-------|------|
| alpha    | 13.05 | 0       | 0  | 13.04 | 13.05 | 13.05 | 13.05 | 13.05 | 14731 | 1    |
| beta[1]  | 0.12  | 0       | 0  | 0.11  | 0.12  | 0.12  | 0.12  | 0.13  | 6526  | 1    |
| beta[2]  | 0.01  | 0       | 0  | 0.00  | 0.01  | 0.01  | 0.01  | 0.01  | 11351 | 1    |
| beta[3]  | 0.07  | 0       | 0  | 0.06  | 0.07  | 0.07  | 0.07  | 0.07  | 9726  | 1    |
| beta[4]  | -0.02 | 0       | 0  | -0.02 | -0.02 | -0.02 | -0.02 | -0.01 | 11173 | 1    |
| beta[5]  | 0.16  | 0       | 0  | 0.16  | 0.16  | 0.16  | 0.16  | 0.17  | 13351 | 1    |
| beta[6]  | -0.03 | 0       | 0  | -0.03 | -0.03 | -0.03 | -0.02 | -0.02 | 14086 | 1    |
| beta[7]  | 0.09  | 0       | 0  | 0.08  | 0.09  | 0.09  | 0.09  | 0.09  | 8872  | 1    |
| beta[8]  | 0.02  | 0       | 0  | 0.02  | 0.02  | 0.02  | 0.02  | 0.03  | 17872 | 1    |
| beta[9]  | 0.26  | 0       | 0  | 0.25  | 0.25  | 0.26  | 0.26  | 0.26  | 8396  | 1    |
| beta[10] | 0.04  | 0       | 0  | 0.04  | 0.04  | 0.04  | 0.04  | 0.05  | 20022 | 1    |
| sigma    | 0.31  | 0       | 0  | 0.31  | 0.31  | 0.31  | 0.32  | 0.32  | 8190  | 1    |

## II. References

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